

Predicting Twitter Virality Using Topic Modeling, Linguistic Features, and Machine Learning

Abstract

Predicting which social media posts become viral is challenging due to the complex relationships between language, topic, and user characteristics. This study investigates whether tweet virality can be predicted using linguistic features, author metadata, and topic modeling. Tweets from a publicly available March 2024 Twitter dataset were preprocessed and used to extract features describing writing style, readability, sentiment, emotion, and user characteristics. BERTopic was applied to identify discussion topics, which were consolidated into broader thematic categories for statistical analysis. Mann-Whitney U tests were used to compare sentiment distributions between viral and non-viral tweets within each topic. Three predictive models—Logistic Regression, Random Forest, and BERTweet—were evaluated using Precision, Recall, F1-score, and ROC-AUC. Model interpretability was examined using Logistic Regression coefficients, Random Forest feature importance, and SHAP values. This work demonstrates how combining topic modeling with interpretable machine learning provides insight into the factors associated with social media engagement.

1. Introduction

Social media platforms generate large volumes of user-created content, but only a small percentage of posts receive substantial engagement. Understanding why certain tweets become viral has applications in marketing, journalism, public communication, and computational social science.

Previous work has shown that user metadata, network characteristics, and textual content all influence engagement. However, fewer studies have examined how linguistic characteristics vary across different discussion topics. This project investigates whether combining linguistic features, author metadata, and topic modeling improves both the prediction and interpretation of tweet virality.

The study addresses three research questions:

1. Which linguistic characteristics are associated with tweet virality?

2. Do these relationships differ across discussion topics?
3. How do traditional machine learning models compare with transformer-based language models for predicting virality?

To answer these questions, tweets were analyzed using BERTopic, statistical hypothesis testing, and supervised machine learning models.

2. Methodology

2.1 Dataset and Preprocessing

The study used the Twitter New Dataset (March 2024) obtained from Kaggle. Tweets were cleaned by removing special characters, normalizing whitespace, and lemmatizing text. An engagement score was calculated using likes, retweets, and replies, then normalized by follower count to compute an engagement rate. Tweets within the top 10% of engagement rate were labeled as viral, creating a binary classification problem.

2.2 Feature Engineering

To capture multiple aspects of tweet content, several groups of features were extracted:

- **Author metadata:** follower count, following count, and verification status.
- **Linguistic features:** text length, punctuation usage, hashtags, mentions, and URLs.
- **Readability:** Flesch Reading Ease, Flesch-Kincaid Grade Level, and Gunning Fog Index.
- **Sentiment and subjectivity:** VADER compound sentiment and TextBlob polarity and subjectivity.
- **Emotion:** transformer-based emotion classification.
- **Topic information:** BERTopic topic assignments represented using one-hot encoding.

These features served as inputs for the supervised machine learning models.

2.3 Topic Modeling

BERTopic was applied using OpenAI **text-embedding-3-small** embeddings to identify semantically related tweet clusters. Representative keywords and sample tweets were manually reviewed to assign descriptive labels. To improve interpretability and increase sample sizes for downstream analyses, the original topics were grouped into broader categories:

- International Affairs & Security
- Government & Public Policy
- Domestic Politics & Society
- Community & Public Services
- Other / Miscellaneous

These generalized topics were used for statistical analysis and predictive modeling.

2.4 Statistical Analysis

To determine whether sentiment differed between viral and non-viral tweets, Mann-Whitney U tests were performed within each generalized topic using VADER compound sentiment scores. Violin plots were created to visualize sentiment distributions and complement the statistical tests.

2.5 Predictive Modeling

Three supervised learning models were evaluated:

- **Logistic Regression** as an interpretable baseline.
- **Random Forest** to capture nonlinear relationships and estimate feature importance.
- **BERTweet**, a transformer model pretrained on Twitter text, to evaluate prediction directly from raw tweet content.

Models were evaluated using Precision, Recall, F1-score, and ROC-AUC.

3. Results

Model performance is summarized in **Table X**.

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.1933	0.7910	0.3107	0.8018
Random Forest	0.9667	0.2636	0.4143	0.8862

Statistical analysis showed that sentiment differed significantly between viral and non-viral tweets for several topic categories, suggesting that the relationship between sentiment and engagement depends on discussion topic. Feature importance analysis further demonstrated

that author metadata, linguistic features, and topical information all contributed to prediction performance, although their relative importance varied across models.

4. Discussion

This study demonstrates that tweet virality can be analyzed through a combination of linguistic feature engineering, topic modeling, and machine learning. BERTopic provided meaningful thematic groupings that enabled topic-specific statistical analysis, while feature importance methods identified the characteristics most strongly associated with engagement.

The comparison of Logistic Regression, Random Forest, and BERTweet illustrates the trade-off between interpretability and predictive performance. Traditional models provide transparent explanations through coefficients and feature importance, whereas BERTweet learns contextual representations directly from tweet text. Together, these approaches offer both predictive capability and insight into the factors influencing social media engagement.

5. Limitations and Future Work

This study is limited to English-language tweets collected during a single month and does not consider multimedia content, posting time, or network diffusion. Future work could incorporate multimodal information, larger datasets spanning longer time periods, and additional transformer architectures. Topic-specific predictive models may also provide further insight into how engagement varies across different discussion themes.

6. Conclusion

This project developed a framework for predicting Twitter virality using linguistic features, author metadata, BERTopic topic modeling, and supervised machine learning. By combining statistical analysis with interpretable and transformer-based models, the study provides insight into the linguistic and topical factors associated with online engagement. The results demonstrate the value of integrating topic modeling with predictive machine learning to better understand social media behavior.

References

1. Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*.
2. Nguyen, D. Q., Vu, T., & Nguyen, A. T. (2020). *BERTweet: A pre-trained language model for English Tweets*.
3. Hutto, C. J., & Gilbert, E. (2014). *VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text*.
4. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*.
5. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*.
6. Kaggle. *Twitter New Dataset (March 2024)*.